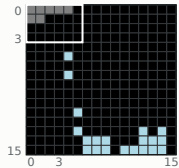


Learning to distinguish touching objects in Ice

Four game frames, evaluated by the same learned P and B at each checkpoint.

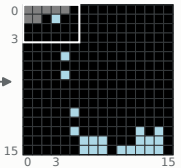
Gameplay →

X₀ · step 205



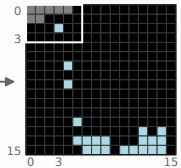
Touching gray objects

X₁ · step 206



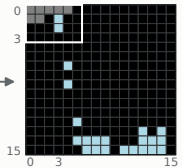
A drop appears at (1,3)

X₂ · step 207



The drop falls to (2,3)

X₃ · step 208



Another drop at (1,3)

Learning ↓

blue: added · orange: error · green: repair · gray: unchanged

Iteration 3 · node 3

P lists colored cells

P(X₀)

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(5,4),
lightblue(7,4),...
```

P(X₁)

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(1,3),
lightblue(5,4),...
```

P(X₂)

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(2,3),
lightblue(6,4),...
```

P(X₃)

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(1,3),
lightblue(2,3),...
```

min(ID, cFD) = 0.100

B = ∅ (no world knowledge)

Iteration 8 · node 6

B names objects and their dynamics

P(X₀)

unchanged

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(5,4),
lightblue(7,4),...
```

P(X₁)

unchanged

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(1,3),
lightblue(5,4),...
```

P(X₂)

unchanged

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(2,3),
lightblue(6,4),...
```

P(X₃)

unchanged

```
bg=black;
gray(0,0),gray(0,1),...
gray(0,4),gray(1,0),
gray(1,1),
lightblue(1,3),
lightblue(2,3),...
```

min(ID, cFD) = 0.583

B — updated since iteration 3

```
... fixed 2x2 block at cells (0,0),(0,1),(1,0),(1,1). ...
... a movable block: three contiguous gray cells on row 0. ...
On noop: every blue/lightblue cell not on row 15 attempts to move down one row. ...
On down: if the cell (1, c) is empty ... where c is the middle column of the movable block ...
a new blue/lightblue cell appears at (1, c) ...
```

Iteration 14 · node 8

P introduces object roles, but merges the gray objects

P(X₀)

```
... fixed:gray(0,0),...
movable:gray_row0:
0,1,2,3,4;
other:gray_row0:2-4;
lightblue_row5:4;
lightblue_row7:4;...
```

P(X₁)

```
... fixed:gray(0,0),...
movable:gray_row0:
0,1,2,3,4;
other:gray_row0:2-4;
lightblue_row1:3;
lightblue_row5:4;...
```

P(X₂)

```
... fixed:gray(0,0),...
movable:gray_row0:
0,1,2,3,4;
other:gray_row0:2-4;
lightblue_row2:3;
lightblue_row6:4;...
```

P(X₃)

```
... fixed:gray(0,0),...
movable:gray_row0:
0,1,2,3,4;
other:gray_row0:2-4;
lightblue_row1:3;
lightblue_row2:3;...
```

min(ID, cFD) = 0.602

B — unchanged from iteration 8 (shortened excerpt)

```
... a movable block: three contiguous gray cells on row 0. ...
On down: if the cell (1, c) is empty ... where c is the middle column of the movable block ...
a new blue/lightblue cell appears at (1, c) ...
```

Iteration 22 · node 12

P separates the fixed object from the three-cell cloud

P(X₀)

```
... fixed:gray(0,0),...
movable:gray_row0:
2-4;
other:lightblue_row5:4;
lightblue_row7:4;...
```

P(X₁)

```
... fixed:gray(0,0),...
movable:gray_row0:
2-4;
other:lightblue_row1:3;
lightblue_row5:4;...
```

P(X₂)

```
... fixed:gray(0,0),...
movable:gray_row0:
2-4;
other:lightblue_row2:3;
lightblue_row6:4;...
```

P(X₃)

```
... fixed:gray(0,0),...
movable:gray_row0:
2-4;
other:lightblue_row1:3;
lightblue_row2:3;...
```

min(ID, cFD) = 0.725

B — unchanged from iteration 8 (shortened excerpt)

```
... a movable block: three contiguous gray cells on row 0. ...
On down: if the cell (1, c) is empty ... where c is the middle column of the movable block ...
a new blue/lightblue cell appears at (1, c) ...
```